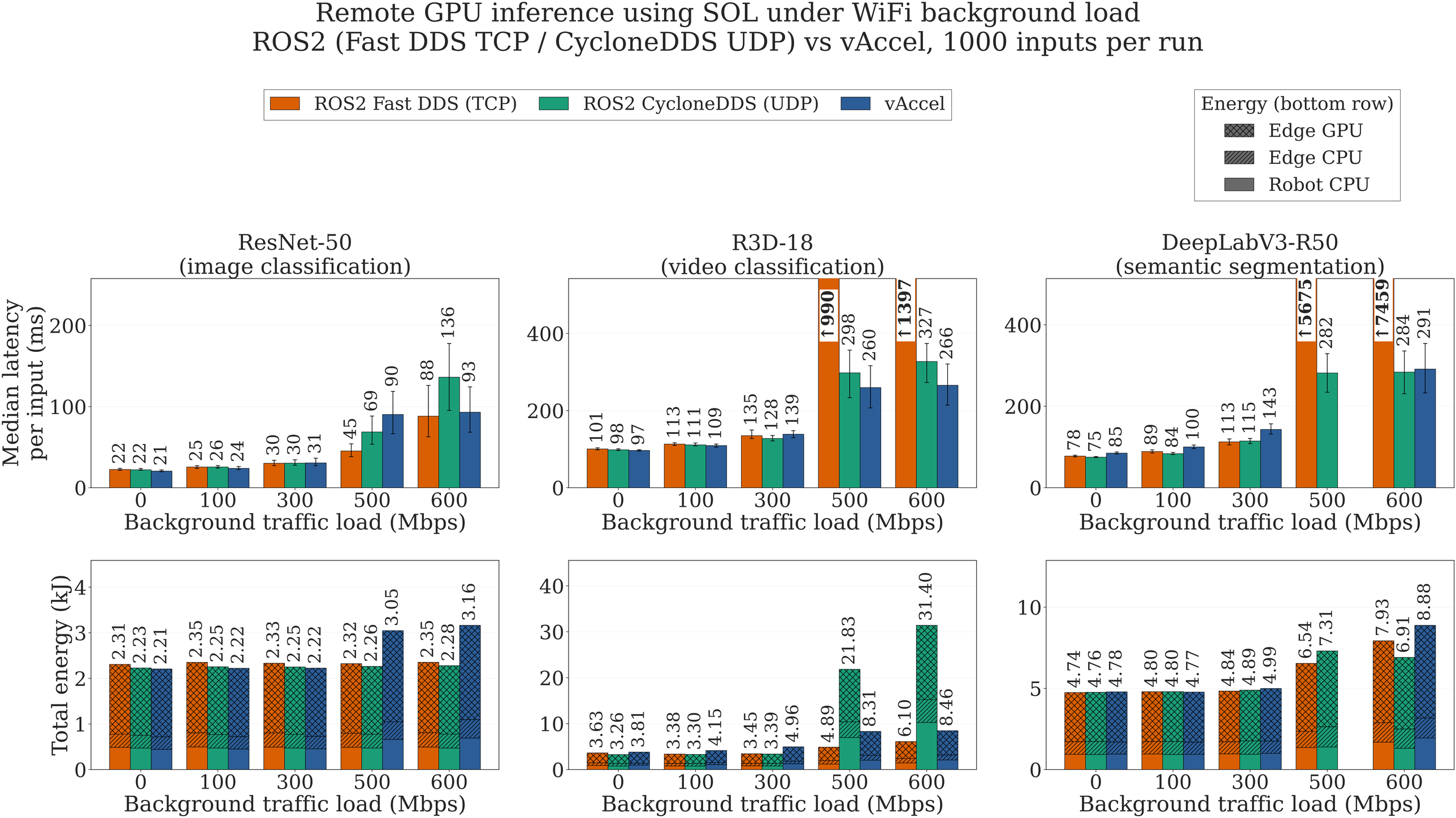
Remote GPU inference using SOL under WiFi background load
ROS2 (Fast DDS TCP / CycloneDDS UDP) vs vAccel, 1000 inputs per run
ROS2 Fast DDS (TCP)
ROS2 CycloneDDS (UDP)
vAccel
Energy (bottom row)
Edge GPU
Edge CPU
Robot CPU
ResNet-50
(image classification)
R3D-18
(video classification)
DeepLabV3-R50
(semantic segmentation)
Median latency per input (ms)
Total energy (kJ)
Background traffic load (Mbps)
0
100
300
500
600
22 22 21
25 26 24
30 30 31
45 69 90
88 136 93
101 98 97
113 111 109
135 128 139
↑990 298 260
↑1397 327 266
78 75 85
89 84 100
113 115 143
↑5675 282
↑7459 284 291
2.31 2.23 2.21
2.35 2.25 2.22
2.33 2.25 2.22
2.32 2.26 3.05
2.35 2.28 3.16
3.63 3.26 3.81
3.38 3.30 4.15
3.45 3.39 4.96
4.89 21.83 8.31
6.10 31.40 8.46
4.74 4.76 4.78
4.80 4.80 4.77
4.84 4.89 4.99
6.54 7.31
7.93 6.91 8.88